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SPECIFICATION FOR TOOL

An innovative platform for global clinical trial discovery

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Prepared By: Ahamed Kabeer

Website
<https://www.iclinicalai.cloud/>

Product Overview Section

Our company is dedicated to developing innovative healthcare solutions. We aim to enhance clinical research accessibility and efficiency through our AI-Enabled Clinical Research Search Tool, enabling stakeholders to discover and filter clinical trials globally while integrating both structured and unstructured data seamlessly.

By harnessing the power of Natural Language Processing (NLP) and advanced filtering techniques, we create an intuitive user experience tailored for patients, clinicians, researchers, and CROs. Our platform transforms the way clinical trials are analyzed, ensuring that vital information is easily accessible and actionable in real-time.

Organization and Management



Ahamed Kabeer
Program Director



Ilayaraja Veluchamy
Head - IT Administration



Varun Kumar
Manager - Clinical Trial
& Data Management



Avijith Das
Architech - AI and ML

MISSION AND VISION

Revolutionizing access to clinical research

Mission

Our mission is to enhance **clinical research access** for all stakeholders.

Vision

Our vision is to create a world where **innovation** in clinical trials leads to better health outcomes for patients globally.



GOALS OF THE PRODUCT

30%

Engage patients effectively in trials

50%

Enhance research efficiency for stakeholders

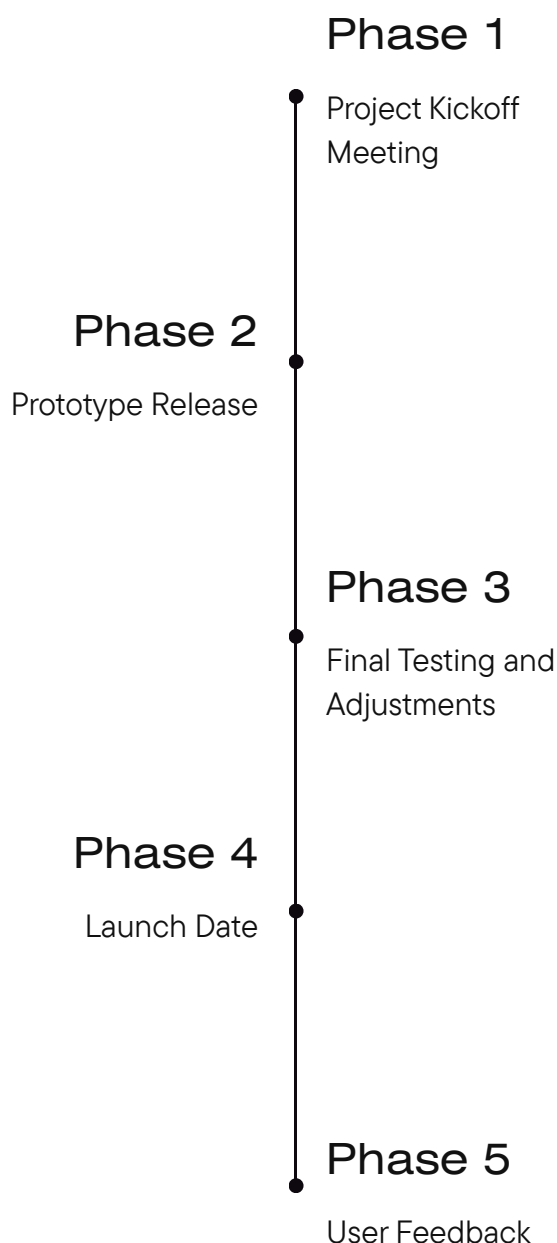
10%

Streamline clinical trial discovery process

The primary goal of the AI-Enabled Clinical Research Search Tool is to **enhance research efficiency** and foster patient engagement in clinical trials. By creating an advanced digital platform that integrates **structured and unstructured data**, our tool will empower stakeholders—patients, clinicians, researchers, and CROs—to seamlessly discover, filter, and analyze global clinical trials. With intuitive user experience (UX) design and advanced natural language processing (NLP), the tool will bridge the information gap in clinical research. It aims to facilitate informed decision-making and improve trial recruitment by allowing users to easily identify relevant trials based on specific criteria. In doing so, we aspire to not only streamline the research process but also enhance patient outcomes through increased participation in suitable clinical studies.

DEVELOPMENT TIMELINE

The timeline outlines key milestones for the development and implementation of the AI-Enabled Clinical Research Search Tool. It begins with the **Project Kickoff Meeting**, setting clear objectives and team roles. Following this, we will create a functional **Prototype Release** allowing for initial testing. The project will reach the **Final Testing and Adjustments** phase, ensuring the tool meets all criteria before the **Launch Date**. Post-launch, we will gather **User Feedback** to identify areas for improvement and enhancements based on stakeholder experiences and needs.



FEATURE SPECIFICATION

Core Search & Filtering Capabilities

Keyword Search

- Basic search by trial title, condition, intervention, sponsor, investigator, location.
- Advanced search with Boolean operators (AND, OR, NOT) and phrase recognition.

Disease/Condition Specific Search

- Search using ICD-10 codes, SNOMED CT.
- Filter by disease severity, stage, biomarkers.

Intervention/Drug Specific Search

- Search by drug name (generic/brand), class, mechanism, intervention type.
- Filter by dosage, frequency, route of administration.

Trial Status Filtering

- Recruiting, Not yet recruiting, Active, Suspended, Terminated, Completed, Withdrawn, Unknown.

Phase Filtering

- Phase 0, I, II, III, IV, Early Phase 1, Not Applicable.

Study Type Filtering

- Interventional, Observational, Expanded Access, Other

FEATURE SPECIFICATION

Core Search & Filtering Capabilities

Eligibility Criteria Filtering

- Age range, Gender, Inclusion/Exclusion criteria

Outcome Measures Filtering

- Primary, Secondary

Sponsor/Funder Filtering

- Filter by type: Pharma, Academic, Government

Date Filtering

- Start date, Completion date, Last update

Location Filtering

- Country, State, City, Institution.
- Geographic radius based on postal code/address.

Study Design Filtering

- RCTs, Open-label, Single-blind, Double-blind, Crossover, etc.

FEATURE SPECIFICATION

User Interface & Experience Features

Search Interface

- Clean UI with intuitive auto-suggest.
- Saved search history and favorite filters.

Refinement Options

- Real-time dynamic filters.
- Faceted navigation.

Results Display

- Summary and detailed views.
- Sorting by relevance, date, proximity.

Trial Comparison Tool

- Side-by-side feature comparison of selected trials

Interactive Maps

- Pin-based map interface showing site locations

User Accounts & Personalization

- Save trials, set alerts, receive recommendations

Accessibility

- Screen reader support, keyboard navigation, high contrast mode

Multi-language Support

- Interface and summaries in multiple languages

FEATURE SPECIFICATION

Data & Information Presentation

Trial Details

- NCT Number, Title, Summary, Description.
- Eligibility, Contact Info, Locations, Sponsors.
- Study Arms, Outcome Measures, Design.
- Links to registries, publications.

Visualizations

- Graphs for recruitment trends, location heatmaps

External Integration

- PubMed, DrugBank, FDA links

Lay Summaries

- Patient-friendly explanations of trials

Export Options

- SV, PDF, JSON formats for data exports

FEATURE SPECIFICATION

Advanced & Value-Added Features

Patient Matching Algorithm

- Match patient profiles to relevant trials

Educational Resources

- Info on trial participation, ethics, consent

Investigator/Site Profiles

- Portfolio and history of research activity.

Developer API

- RESTful API for external data consumption

Trial Analytics & Trends Dashboard

- Visual analytics on trial types, sponsors, regions, growth

Mobile App

- Optimized for iOS and Android.

Community Forum

- Discussion platform for patients, caregivers, clinicians

NLP and AI Features

- Intelligent query understanding, keyword expansion, auto-tagging.

Feedback Mechanism

- User feedback collection for continuous improvement.

FEATURE SPECIFICATION

Backend & Technical Infrastructure

Data Aggregation

- APIs and crawlers for ClinicalTrials.gov, WHO ICTRP, EU CT, etc.
- Regular synchronization and deduplication.

Architecture

- Scalable microservices.
- Cloud-native deployment.

Security & Compliance

- GDPR, HIPAA compliance.
- Encrypted data storage and transmission.

Performance Optimization

- Fast indexing, cached search results.

API Access

- Access-controlled API endpoints for partners.

DEVELOPMENT ROADMAP

Phase 1

Core search, trial
data integration,
basic UI.

Phase 3

AI/NLP layer, patient
matching, mobile
app.

Phase 2

Advanced filtering,
personalization,
visualization.

Phase 4

External APIs, community
features, analytics
dashboard.

Launch

TARGET USERS

Clinical
researchers,
CROs, pharma
companies

Physicians,
clinicians &
professional
involved in
clinical trials

Patients and
caregivers

Regulatory
bodies and
academic
institutions



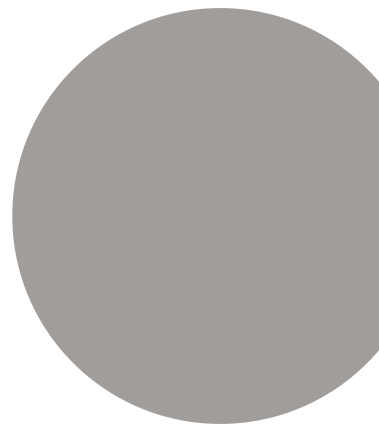
METRICS FOR SUCCESS

- Search speed and relevance.
- User engagement and retention.
- Trial match success rate.
- Data accuracy and freshness.

APPENDICES

- Search speed and relevance.
- User engagement and retention.
- Trial match success rate.
- Data accuracy and freshness.

CONTACT INFORMATION: LET'S CONNECT!



iCLINICAL.AI
P O W E R E D B Y I D D C R

Phone

+91 770 222 77 00

Email

info@iclinicalai.cloud

Website

www.iclinicalai.cloud

